DAY 08: HANDS-ON EC2 PROVISIONING & DOCKER IMPLEMENTATION

Cloud Computing & DevOps Master Curriculum

1. Setting Up Docker on Amazon Linux 2 / Ubuntu EC2

Step-by-step commands to install, run, and configure Docker on an EC2 instance:

```
# Update packages & install Docker
sudo yum update -y
sudo amazon-linux-extras install docker -y   # On Amazon Linux 2
# OR for Ubuntu: sudo apt install -y docker.io

# Start and enable Docker service
sudo systemctl start docker
sudo systemctl enable docker

# Add ec2-user to docker group (run docker without sudo)
sudo usermod -aG docker ec2-user
newgrp docker
```

2. Deploying Application Containers on EC2

Running an Nginx Web Server container with port mapping:

```
# Pull and run Nginx container
docker run -d --name my-nginx -p 80:80 nginx:latest

# Check running containers
docker ps

# View container logs
docker logs my-nginx

# Execute interactive shell inside container
docker exec -it my-nginx /bin/bash
```

3. Security Group Port Binding

To access the Nginx web application running inside Docker from your web browser, ensure the EC2 Security Group inbound rule allows **HTTP (Port 80)** traffic from **0.0.0.0/0**.

KEY TAKEAWAYS & SUMMARY

✔ Adding users to the 'docker' group allows running docker commands without root `sudo` privileges.

✔ Port mapping (-p HostPort:ContainerPort) bridges external traffic into the isolated container network.

✔ Docker ps and docker logs are primary tools for monitoring container health and troubleshooting.